Ultrasonics 54 (2014) 614–620

Contents lists available at ScienceDirect

Ultrasonics

journal homepage: www.elsevier.com/locate/ultras

Microfabrication of stacks of acoustic matching layers for 15 MHz ultrasonic transducers

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ARTICLE INFO

Article history:
Received 10 May 2013
Received in revised form 24 August 2013
Accepted 24 August 2013
Available online 30 August 2013

Keywords:
High frequency transducers
Silicon–polymer composite
Silicon micromachining
Matching layer

ABSTRACT

This paper presents a novel method used to manufacture stacks of multiple matching layers for 15 MHz piezoelectric ultrasonic transducers, using fabrication technology derived from the MEMS industry. The acoustic matching layers were made on a silicon wafer substrate using micromachining techniques, i.e., lithography and etch, to design silicon and polymer layers with the desired acoustic properties. Two matching layer configurations were tested: a double layer structure consisting of a silicon–polymer composite and polymer and a triple layer structure consisting of silicon, composite, and polymer. The composite is a biphase material of silicon and polymer in 2-2 connectivity. The matching layers were manufactured by anisotropic wet etch of a (1 1 0)-oriented Silicon-on-Insulator wafer. The wafer was etched by KOH 40 wt%, to form 83 μm deep and 4.5 mm long trenches that were subsequently filled with Spurr's epoxy, which has acoustic impedance 2.4 MRayl. This resulted in a stack of three layers: The silicon substrate, a silicon–polymer composite intermediate layer, and a polymer layer on the top. The stacks were bonded to PZT disks to form acoustic transducers and the acoustic performance of the fabricated transducers was tested in a pulse-echo setup, where center frequency, −6 dB relative bandwidth and insertion loss were measured. The transducer with two matching layers was measured to have a relative bandwidth of 70%, two-way insertion loss 18.4 dB and pulse length 196 ns. The transducers with three matching layers had fractional bandwidths from 90% to 93%, two-way insertion loss ranging from 18.3 to 25.4 dB, and pulse lengths 326 and 446 ns. The long pulse lengths of the transducers with three matching layers were attributed to ripple in the passband.

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1. Introduction

Ultrasonic investigation of microstructures in soft tissues, e.g. composition of small tumors or a vessel wall, requires an imaging system that can provide high resolution in both the lateral and the axial direction. Such a system requires a high frequency (10–100 MHz) broadband (>50% −6 dB relative bandwidth) ultrasound transducer, as the axial resolution is dependent on the bandwidth and the lateral resolution is inversely proportional to the center frequency of the transducer [1]. There are two main challenges associated with such a transducer: Obtaining the correct thickness of the active layer and matching the acoustic impedance of the active layer to that of human tissue. *Most current ultrasonic transducers are based on a ferroelectric active layer, normally working in thickness mode*. To operate in the frequency range from 10 to 100 MHz, the thickness of the active layer must be between 20 and 200 μm. This is difficult and expensive to achieve by conventional fabrication methods, e.g. lapping [2]. Moreover, the acoustic impedance of the active layer, e.g. 20–40 MRayl for a ceramic, or 15–20 MRayl for a piezocomposite, is much higher than that of the human tissue, 1.6 MRayl. To enhance transmission of acoustic energy from the piezoelectric element into the tissue, acoustic matching layers with acoustic impedances between that of the active layer and the load material are required [3–6]. Several approaches are available for designing the optimal acoustic matching [3–5], but finding single materials with optimal acoustic impedances for these configurations are in general not possible, and composites are often used [6]. Typical composites used as matching layers are mixtures of metal particles in a polymer matrix. However, employing these materials at high frequencies, e.g. 10–100 MHz, introduces some challenges such as non-uniformity, high attenuation and dispersion [6].

We have previously demonstrated a novel design for acoustic matching layers in high frequency broadband ultrasound transducers using silicon micromachining, i.e. lithography and etch, to fabricate composite acoustic matching layers [7]. In these designs, the active piezoelectric layer can be manufactured by thick-film technology, which is able to produce the ceramic layer in the range of interest (20–200 μm). The ceramic film can be made either by

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http://dx.doi.org/10.1016/j.ultras.2013.08.015

tape casting methods [8] and subsequently bonded to a silicon substrate, or by screen-printing or spin-coating directly on a platinum-coated silicon substrate [2,9,10]. The silicon substrate is micromachined to build a stack of multiple matching layers with descending acoustic impedance from the piezoelectric layer to the tissue, while the transducer is left air backed. Instead of performing alternately deposition and lapping layer by layer as in conventional transducer manufacturing, the proposed method allows us to build a stack of multiple layers in one process. This is a great advantage when the transducers become thin, e.g. less than 100 μm. These fabrication methods are well established for mass production in the MEMS industry, where a batch-production process is used to fabricate multiple identical devices on a single silicon wafer. If these methods can be applied to ultrasound transducer production, it may open for low-cost mass-production of high frequency transducers, suitable for e.g. disposable medical devices.

This paper addresses one of the mentioned issues, i.e. the manufacturing of stacks of acoustic matching layers using silicon micromachining techniques. Following the initial work of Haller and Khuri-Yakub on micromachined composites [11], the aim of this study is to develop multiple micromachined matching layers for high frequency (10–100 MHz) ultrasonic transducers using this composite as one of the matching layer. Two configurations of matching layers were tested: A two layer design, where the inner layer is a silicon–polymer composite and the outer layer is a polymer, and a three layer design, where the inner layer is silicon, the intermediate layer is a silicon–polymer composite and the outer layer is a polymer. The micromachined silicon–polymer composite was made as 2-2 connectivity, i.e. consisting of long, narrow lines of alternately silicon and polymer. This composite was fabricated by etching deep, long trenches into the silicon substrate, and these grooves were subsequently filled with epoxy resin [11]. For testing the performance of the acoustic matching configurations, three transducer prototypes were fabricated for operating at 15 MHz. At this frequency, we have control of fabrication parameters and uncertainties [7]. The performance was tested by pulse-echo measurements in a water tank. Center frequency, bandwidth and insertion loss were measured. The measure curves were compared with simulation results from a one dimensional equivalent circuit model based on the Mason model [12], which has been shown to give identical results as KLM model [13].

2. Materials and methods

2.1. Design of the acoustic impedance matching

Acoustical matching the high impedance of the piezoelectric material (~37 MRayl) to the relatively low impedance of human tissue (~1.6 MRayl) is challenging. For the two matching layer configuration, we followed the guidelines recommended by Desilets et al. [3] to achieve a maximally flat response. According to this model, the optimal acoustic impedance values for the inner and outer matching layers are 9.5 and 2.4 MRayl (Table 1). Spurr's epoxy (Low viscosity embedding media Spurr's Kit, Electron Microscopy Sciences, Hatfield, PA, USA) is a material that has low viscosity (60 cP), and small heat shrinkage [14], making it well-suited to fill the etched grooves without damaging the silicon posts [15]. Equally important for this application, its acoustic impedance is 2.4 MRayl, i.e. ideal for the outer acoustic matching layer. To achieve a middle layer with acoustic impedance 9.5 MRayl, a silicon–polymer composite of Spurr's epoxy with 35% volume fraction silicon is fabricated, calculated from the dynamic model of guided waves [16]. Due to the risk of cracking the composite along the silicon crystal orientations during the grinding process, a thin layer of silicon, 10 μm thick, was kept underneath the composite layer. Silicon has high longitudinal sound velocity (8440 m/s) and lov mechanical loss (loss tangent 0.0025 [15]), and this layer has al most no influence on the transducer performance. 1D simulation with and without this silicon layer predict less than 1% differenc in both bandwidth and sensitivity.

For the triple matching layer configuration, we used a differen set of acoustic impedance values, shown in Table 1. Starting at th inner layer, close to the PZT disc, and proceeding outwards, the lay ers consist of silicon (19.7 MRayl), 2-2 composite with 24% silico volume fraction (7.6 MRayl) and Spurr's epoxy (2.4 MRayl). Thes values were designed to approximate a Chebyshev bandshap transfer function, which ideally consists of layers with impedance 24.9, 7.6 and 2.3 MRayl [17].

2.2. Manufacturing process

The fabrication process illustrated in Fig. 1 was used to build th transducers. First, a 0.5 μm thick layer of thermal silicon dioxide SiO_2, was grown on top of a 100 mm diameter Silicon-On-Insulato SOI, wafer (Ultrasil Corporation, USA). The SOI wafer consist of a (83 ± 1) μm thick (1 1 0)-oriented Si device layer, a 0.5 μm thic buried SiO_2 layer, and a (500 ± 10) μm thick (1 0 0)-oriented Si han dle layer (Fig. 1a). The (1 1 0)-orientation of the Si device layer wa chosen to allow use of anisotropic wet–etch to form deep, long vertical sidewall trenches along the {1 1 1} crystal planes, as show by Haller and Khuri-Yakhub [11]. In contrast, anisotropic wet etc on the more common (1 0 0)-oriented Si wafers creates structure with sidewalls sloped at angle 54.7°. The buried SiO_2 layer will sto the etching at a thickness defined by the device layer, (83 ± 1) μm A 1.5 μm thick layer of positive photoresist (S1813, Shipley Com pany Inc., Boston MA, USA) was spin coated on top of the oxid layer. A glass mask (The International Phototool Company, LL CO, USA) with parallel line features, tolerance ±0.3 μm, was usec These features were transferred to the photoresist using photoli thography with contact mask exposure. The feature lines wer aligned to the {1 1 1} crystal planes of the silicon wafer with a accuracy of 0.1°, using a fan-shaped structure and the method de scribed in [18]. By aligning the lines along the crystal planes in th silicon, anisotropic etch will create trenches with vertical side walls. The SiO_2 was first patterned in buffered oxide etch, BO 7:1, and the patterned SiO_2 then served as mask for the anisotropi wet–etch of the silicon device layer (Fig. 1b). Silicon wet etch wa performed with KOH 40 wt% etchant at 70 °C for 100 min, formin deep grooves into the substrate (Fig. 1c). The SiO_2 mask was the stripped in BOE 7:1, the wafer was rinsed in de-ionized wate and air dried. The grooves in the etched silicon wafer were fille by Spurr's epoxy, and cured at 70 °C for 8 h to form a stack of sili con, composite and polymer (Fig. 1d).

The processed wafers were diced into 6 mm × 6 mm dies usin a Disco Abrasive Systems dicing saw (Disco Corporation, Tokyo, Ja pan). The top polymer and bottom silicon layers of each sampl were ground to their target thicknesses and their surfaces wer polished. The lapping process was done using MultiPrep™ Syster grinding and polishing equipment (Allied High Tech Products Inc CA, USA) with a coarse to fine grit scheme with accuracy of 1 μm i thickness and 0.01° in angular positioning. The final lapping parti cle diameter was 5 μm and the polishing chemical was a mixtur of colloidal silica and 0.05 μm alumina (Allied High Tech Product Inc., CA, USA).

The finished stack was glued to a 6 mm × 6 mm PZT plat (*3195HD*, Boston Piezo-optics, Bellingham, MA, USA) using Spurr' epoxy, to form a transducer (Fig. 1e). The roughness was investi gated by an interferometer (*Wyko NT9100 Technical*, Veeco Instru ments Inc., USA), giving an arithmetic mean roughness R_a of abou 0.5–0.6 μm for PZT and 0.2–0.3 μm for the final lapped samples From this, the glue bonding layer thickness was estimated to b

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esign specifications of the transducers in this study. The values given in parentheses are the theoretical values to achieve a transfer function of Chebyshev bandshape, calculated om [17] for acoustic impedance Z_{PZT} = 37.1 MRayl and Z_{load} = 1.54 MRayl. Material properties for Spurr's epoxy and silicon are from [15].

Parameter	Active layer	ML_1[b]	ML_2	ML_3
Two-matching layer	PZT[a]	Composite	Spurr's epoxy	
Acoustic impedance (MRayl)	37.1	9.5	2.4	–
Longitudinal velocity (m/s)	4760	6230	2200	–
Target thickness (μm)	140	83	36	–
Mechanical loss tangent	0.012	0.03	0.03	
Three-matching layer	PZT	Silicon	Composite	Spurr's epoxy
Acoustic impedance (MRayl)	37.1	19.7 (24.9)	7.6 (7.6)	2.4 (2.3)
Longitudinal velocity (m/s)	4760	8440	5400	2200
Target thickness (μm)	140	130	83	36
Mechanical loss tangent	0.012	0.0025	0.03	0.03

[a] PZT: PZT properties were estimated by curve-fitting method.
[b] ML: matching layer.

ess than 1 μm, based upon the assumption that this thickness was efined by the total surface roughness of the PZT and the compos-e [19]. The PZT is (140 ± 1) μm thick. The electrodes were pro-essed to form a coaxial pattern on the back side, to allow lectrical connection from the back side of the disc. This is de-cribed in Section 2.3. The coaxial electrode pattern has a center ircle diameter 3.2 mm, and the gap between the center and the ing is 0.4 mm. *The transducer was mounted on an SMA connector sing conductive epoxy (Epo-Tek EE129-4, Epoxy Technology, Inc., ellerica, MA, USA), similar to the design presented by Foster* [20]. A .12 mm diameter wire connected the transducer electrode to he center connector of the SMA. A Teflon tube was designed to over the SMA and make the transducer waterproof (Fig. 1f).

.3. Piezoelectric active element

A 1.54 cm × 1.54 cm PZT plate with 0.3 μm chrome/gold (Cr/ u) electrodes was processed to form a coaxial electrode pattern, o gain access to the front electrode from the back side. A 1.5 μm ayer of photoresist S1813 was spin-coated on top of the PZT plate nd patterned by standard lithography. The Cr and Au layers were tched using Nickelkromets MC2 (Sunchem AB, Partille, Sweden) nd Guldets 22196 (Sunchem AB). The photoresist was stripped nd the etched patterns were diced into 6 mm × 6 mm dies. A ayer of Cr/Au was sputtered on the sides of the PZT dies to create n electrical connection from the front to the back electrode. Elec-rical impedance of the diced plates, in air, was measured using an IP 8753D Network Analyzer (Agilent Technologies, Santa Clara, A, USA). The impedance measurements were used to estimate he electromechanical properties of the PZT discs in the thickness ode, by fitting the measured impedances to calculations from he Mason model using the Nelder–Mead simplex method [21]. his fitting was done using 201 data points over the frequency ange from 10 to 20 MHz [22].

.4. Silicon–polymer 2-2 composite

The silicon–polymer 2-2 composite is the key layer in our de-igns. The composites have a constant period, 22.5 μm, designed ccording to the guidelines in [22] to keep the polymer width less han a quarter of shear wavelength at the center frequency to avoid ateral resonances. Composites with two volume fractions were ade: Composite A, with 35% silicon volume fraction, was used n the two-layer acoustic matching configuration, while composite , with 24% silicon volume fraction, was used in the three-layer onfiguration. Hence, the target silicon bar widths are 7.9 μm for omposite A and 5.4 μm for composite B. The two composites were esigned and fabricated on the same silicon wafer. The uniformity f the fabricated silicon and polymer composites was evaluated by aking Scanning Electron Microscopy (SEM) cross-sections of ran-domly selected samples from each set. To evaluate the composite properties, two samples from each set were chosen and the silicon and polymer layers were removed by lapping, leaving the composites layer alone. These composites were glued to PZT plates (as described in Section 2.3) using Spurr's epoxy. This gave a set of testing transducers, consisting of a single composite layer on top of a coaxial PZT plate. The thickness of the epoxy bonding layer was not measured, but estimated to less than 1 μm, based upon the assumption that the total surface roughness of the PZT and the composite define the epoxy thickness [19]. The acoustic properties of the composite, i.e. acoustic impedance and longitudinal velocity were estimated from electrical impedance curves measured on these transducers, using the same procedure as for the PZT discs. As the bonding epoxy layer between the PZT disc and the composite is much thinner than the wavelength at the operating frequency, this glue layer is not included in the model used in the fitting. The estimated properties were compared with those calculated from the dynamic model of guided waves [16] and the iso-strain theory [23].

2.5. Transducer characterization

Acoustic pulse-echo measurements were performed in a water tank to investigate the transducers' bandwidth and sensitivity. One two-matching-layer transducer and two three-matching-layer transducers were tested. The transducers were connected to a 50 Ω signal generator (Agilent 33522A, Agilent Technologies, Inc., Santa Clara, CA, USA) via a 2 m RG-58 cable. A single cycle 16 V amplitude sine wave with center frequency f = 15 MHz was used to drive the transducers. Received echo responses were acquired from a polished brass plate, sampled on a digital oscilloscope (WaveSurfer 42Xs, LeCroy Corporation, Chestnut Ridge, NY, USA) and transferred to a PC for further analysis. The brass target was a cylinder with 44 mm diameter and 41 mm thickness, placed at normal incidence to the acoustic axis in the near field zone, less than 1 cm from the transducer surface. The insertion loss was calculated as the ratio between the received echo and the transmitted signal. A fast Fourier transform was performed on the received waveforms to obtain the frequency response. The bandwidth was defined from the frequencies where the power spectrum was reduced 6 dB relative to its peak value, and the pulse length was determined from the duration where the amplitude was reduced 20 dB below its peak value.

3. Results and discussion

3.1. Piezoelectric active element

The measured and curve-fitted impedances of a typical PZT disc operating in air are shown in Fig. 2. These impedance curves were

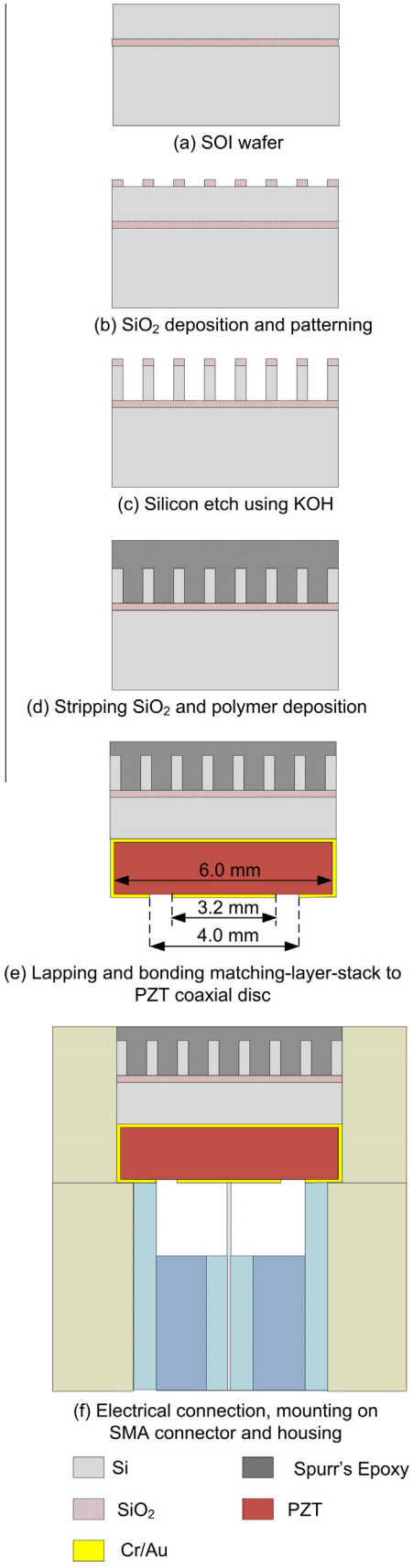


Fig. 1. Main fabrication steps to manufacture multiple micromachined ultrasound transducers.

used to estimate the piezoelectric properties of the disc, resultin in the material properties listed in Table 2. The results are base on measurements on five samples from the same batch. The mate rial data given by the manufacturer are given for comparison.

The electromechanical properties of the fabricated PZT plate differ from the material data given by the manufacturer. This ca be expected, as the manufacturer gives values measured a 1 kHz. Our results are consistent with the parameters from Foste et al. [24], who measured in the frequency range 20–80 MHz.

3.2. Silicon–polymer composite

The polymer kerf and silicon widths for the fabricated compos ite structures are shown in Table 3. The corresponding targe dimensions are given in parentheses for comparison. A SEM imag of the cross-section of a composite of type B is shown in Fig. 3.

Fig. 4 shows the electrical impedance measured on a transduce consisting of a single composite B matching layer bonded on top c a 15 MHz PZT, operating in air. The impedance curve fitted to th Mason model is shown for comparison. These impedance measure ments were used to estimate the acoustic properties of the silicon polymer composite, the resulting parameter estimates are show in Table 4.

The SEM image of the fabricated 2-2 composite, Fig. 3, show vertical sidewalls and flat bottom trenches, suited to be used a one matching layer in multiple-layer stacks. The obtained silico widths matched to the desired dimensions within 5%, as show in Table 3.

In this work, the tolerance on the glass mask was 0.3 μm, caus ing a tolerance in the silicon bar width of 0.5 μm after etching. Th

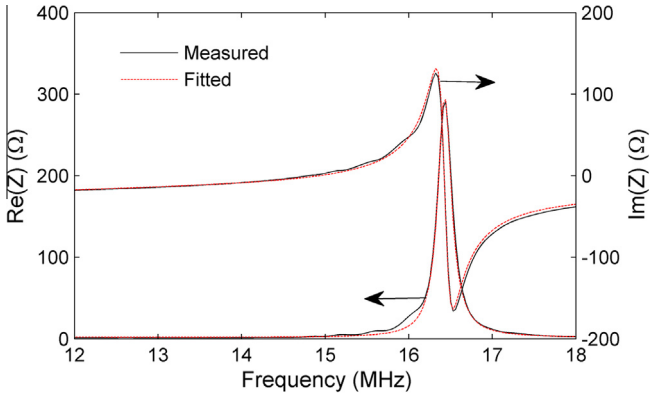


Fig. 2. The measured and curve-fitted impedances of the PZT disc alone in air t estimate the active element of the transducers.

Table 2
The estimated material properties for the proceed PZT coaxial discs.

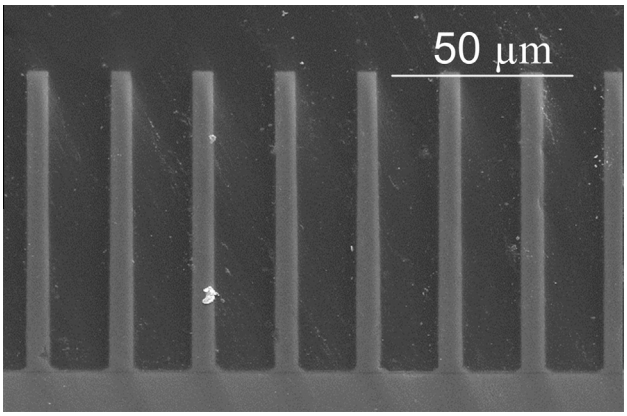
Parameter	Notation	Value	Manufacturer
Density (kg/m^3)	ρ	7800[a]	7800
Elastic stiffness constant at constant electric displacement ($\times 10^{11}$ N/m^2)	c_{33}^D	(1.77 ± 0.25)	1.06
Mechanical loss tangent	$\tan\delta_m$	0.012 ± 0.001	0.0125
Relative permittivity	$\varepsilon^S/\varepsilon_0$	763 ± 69	915
Electrical loss tangent	$\tan\delta_e$	0.078 ± 0.057	0.018
Electromechanical coupling for thickness mode	k_t	0.465 ± 0.005	0.48
Electromechanical coupling loss tangent	$\tan\delta_k$	0.020 ± 0.010	–

[a] From manufacturer.

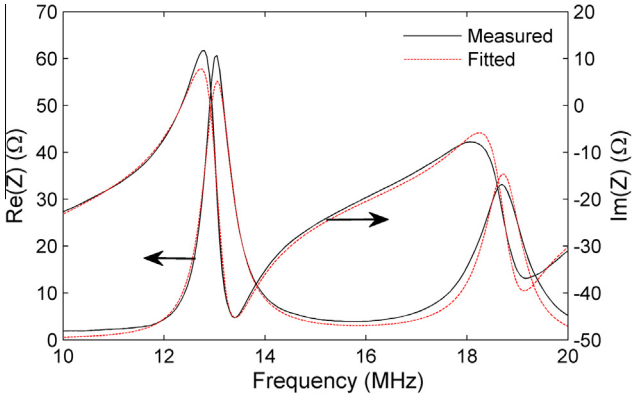
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olymer kerf and silicon width measurements for the fabricated silicon–polymer 2-2 omposite.

Parameter	Dimensions (μm)	
	Composite A	Composite B
Period	22.50	22.50
Polymer kerf width	15.00 (14.6)[a]	16.80 (17.10)
Silicon width	7.50 (7.90)	5.70 (5.40)
Silicon σ (standard deviation)	0.50	0.50

[a] The target design dimensions are depicted between parentheses.



ig. 3. A cross-section image of one of the 2-2 composite B structures fabricated by et etching and used in the three-matching-layer configuration. The bright parts in ie image are silicon, the dark parts are epoxy. The white dots are from the olutions to prepare the sample cross-sections. The composite period is 22.5 μm.



ig. 4. The measured and curve-fitted impedances of one typical transducer ncluding a single composite layer bonded on top of a 15 MHz PZT disc operating in ir. These results were used to estimate the acoustic properties of the composite. his measurement is on a composite from set B with thickness 70 μm.

inimum lateral dimensions of the composite are primarily lim-ed by the features of the mask and of the lithography technology. educing mask tolerance would provide better uniformity. Im-roved mask resolution and more sophisticated lithography, such as electron beam, would make it possible to obtain composites with sub-micron silicon bars [25,26], suitable for being matching layers for transducers operating at higher frequency (above 100 MHz).

The composite material parameters estimated from the impedance measurements are closer to the values calculated from the dynamic model of guided waves than those from the iso-strain model, as mentioned in [14]. The deviations between the estimated values and the values calculated from the dynamic model for velocity and acoustic impedance are 1.4% and 11.8% for composite A and 12.9% and 6.6% for composite B. The difference can be attributed to the difference in the boundary conditions in the dynamic model of guided waves and the experimental structures. The dynamic model was derived under the assumption that the composite was unbounded, whereas in our practical structures, the composite was in between PZT and air. Due to the high acoustic impedance mismatch between the Spurr's epoxy (2.4 MRayl) used for bonding and the composite (7–10 MRayl), the acoustic properties of the complete stack are also sensitive to the bonding thickness [27]. Calculations from the Mason model show the velocity estimate will increase about 10% when the thickness of the epoxy bonding layer increases from 0.1 μm to 1 μm. An alternative approach to estimate the thin composite matching layer might be considered, e.g. the method based on the impedance matching principle by Wang [28]. By transmitting plane waves through a silicon substrate with and without a quarter-wavelength composite layer on top and taking the ratio of these two transfer functions in the frequency domain, one may deduce the velocity of the composite layer at the frequency where this ratio is maximum [28].

3.3. Transducer characterization

The measured pulse-echo results for one two-matching-layer transducer and two three-matching-layer transducers are shown in Figs. 5–7, respectively; together with the simulation results from the Mason model. The received waveforms are shown in the upper graphs, whereas their corresponding spectra are presented in the lower graphs. The spectra were normalized to their maxima for the ease of comparison between simulated and measured results. The comparison between the simulated and measured results is summarized in Table 5.

Pulse-echo measurements on the transducer with two matching layers shows close agreement with the simulations from the Mason model (see Table 5). The −6 dB fractional bandwidth was measured to 70%, the insertion loss to 18.4 dB, and the pulse length to 196 ns. The performance of this transducer is typical for a well-designed transducer with two matching layers, and can also be compared to the $LiNO_3$-based transducer reported by Canata et. al [29], which has center frequency 22 MHz, −6 dB fractional bandwidth of 73%, insertion loss of 19.5 dB and pulse length of 145 ns. This demonstrates the feasibility of using silicon micromachining to manufacture stacks of acoustic matching layers for ultrasound transducers operating at high frequency (10–100 MHz).

Pulse-echo measurements on the three-matching-layer transducers show very large fractional bandwidths, 90–93%, but display long pulse duration, 326–446 ns. The reason for the difference in

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he estimated material properties for silicon–polymer 2-2 composites from curve-fitting the electrical measurements to the mason model. The measured dimensions from able 3 were used to calculate the modeled values.

Parameter	Composite A			Composite B		
	Modeled		Estimated	Modeled		Estimated
	Iso-strain	Dynamic		Iso-strain	Dynamic	
Longitudinal wave velocity (m/s)	6420 ± 138	5900 ± 91	5818 ± 300	5865 ± 173	5609 ± 225	4886 ± 166
Acoustic impedance (MRayl)	9.70 ± 0.40	8.90 ± 0.30	7.85 ± 0.65	8.30 ± 0.40	7.92 ± 0.47	7.40 ± 0.70

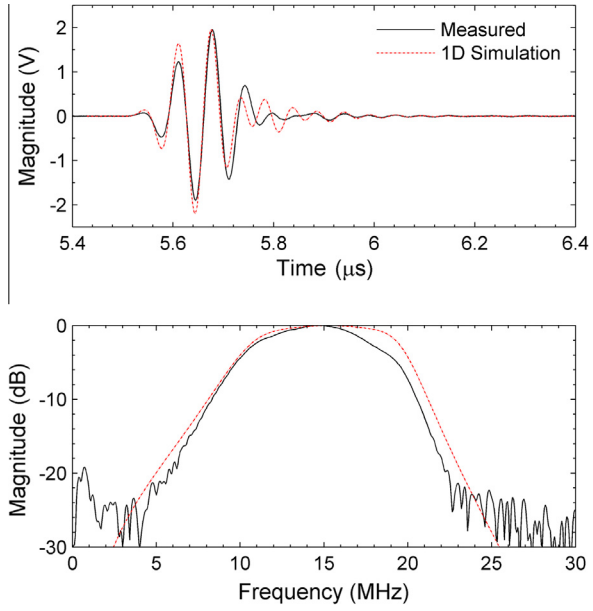


Fig. 5. Pulse-echo responses for a two-matching transducer working at 15 MHz, measured (solid lines) and calculated from the Mason model (dotted lines).

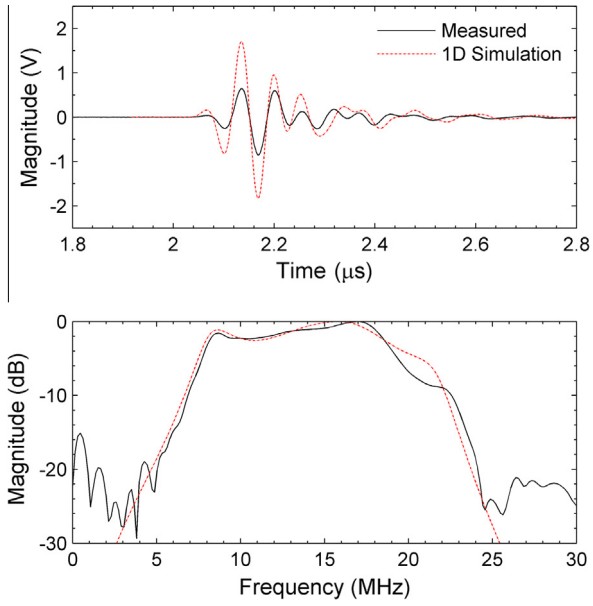


Fig. 6. Pulse-echo responses for a three-matching transducer working at 15 MHz (No. 1), measured (solid lines) and calculated from the Mason model (dotted lines).

echo level between simulated and measured results in one of the transducer measurements was not determined, but it might be attributed to a poor electrical connection between the transducer and the SMA connector. The long pulse durations directly link to the level of ripples in the transducers spectra [5]. The measured

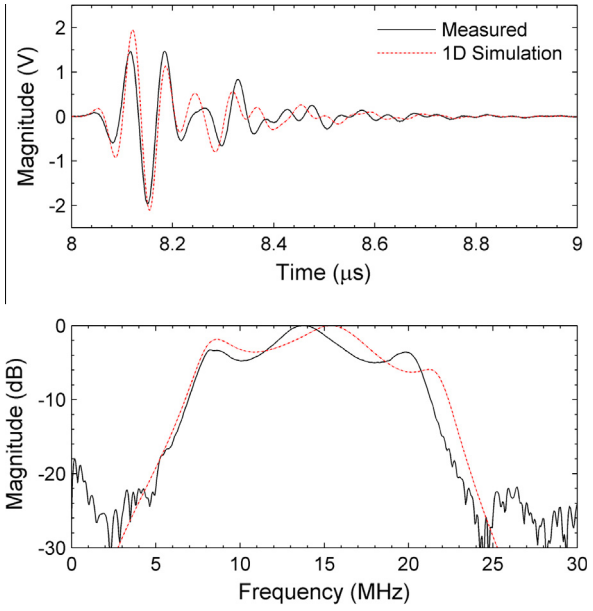


Fig. 7. Pulse-echo responses for a three-matching transducer working at 15 MH (No. 2), measured (solid lines) and calculated from the Mason model (dotted lines

ripple levels for the three-layer transducers were 2.3 and 4.7 dB see Figs. 6 and 7, and less ripple in the transducer spectrum mean a shorter pulse. The performance of this three-matching-layer con figuration can be improved by optimizing the matching layer impedances and thicknesses under constraints to bandwidth, rip ple and loss [5].

Silicon is a crystal with cubic symmetry, and the acousti impedance depends on its orientation. Commercially available sil icon wafers have crystal orientations (100), (110) and (111) wit corresponding acoustic impedances of 19.7, 21.3 and 21.7 MRay respectively [30]. The microfabrication processes presented in thi paper, the anisotropic wet etch; can only be used on (110)-ori ented silicon wafers. When a specific polymer and silicon orienta tion has been chosen, the properties of the intermediate composit layer can be adjusted and optimized by changing the volume frac tion of silicon in the composite material.

Due to the high acoustic impedance mismatch between Spurr' epoxy, PZT and silicon (2.4, 37.1 and 19.7 MRayl) as well as the lov sound velocity of Spurr's epoxy (2200 m/s), the bonding thicknes between the PZT plates and the matching stacks must be kept thir e.g. less than 0.02 λ in our configurations, so that this layer do nc degrade the transducer's performance. Calculations from the Ma son model show that the bonding layer has higher influence o the three-matching-layer configuration than on the two-layer con figuration. According to our calculations, the bonding thicknes could be tolerated to be 2–3 μm for the double matching laye transducers operating at 15 MHz. However, in the triple matchin layer configuration, this layer must be less than 1 μm, to keep th ripple level less than 6 dB in the transducer spectra. Using PZ thick-film printed directly on a platinum-coated silicon wafe [9,10] would eliminate the need for a bonding layer.

Table 5
Comparison between the simulated and measured transducer performance.

Transducer	Mason modeling				Measured results			
	f_c (MHz)[a]	BW (%)	IL (dB)	PL (ns)	f_c (MHz)	BW (%)	IL (dB)	PL (ns)
Two-matching layer	14.9	74.3	17.8	258	14.6	70.2	18.4	196
Three-matching layer No. 1	14.3	97.7	19.2	331	13.6	90	25.4	326
Three-matching layer No. 2	14.0	97.2	18.1	393	14.2	93	18.3	446

[a] f_c is the center frequency; BW is the −6 dB bandwidth; IL is the insertion loss; and PL is the 20 dB pulse length.

. Conclusion

This paper describes how double- and triple-matching-layer onfigurations for transducers operating at 15 MHz have been deigned, built and tested; using silicon micromachining technology rom the MEMS industry. These fabrication methods are wellstablished for mass-production of inexpensive sensors elsewhere. ransducers with large bandwidth (>70%), and low insertion loss <25.5 dB) were obtained. The resulting two-matching-layer transucer showed good performance, with pulse length 196 ns. For the hree-matching-layer, the bandwidth was larger, but the pulses onger. This was attributed to ripple in the passband. Performance f the three-matching-layer transducer can be improved by optiizing the design.

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